UC San Diego

DSC 102 B00
Systems for Scalable Analytics

Spring 2026

Alexander Kolovos

About Myself

Physics major; Crete, Greece.

Ph.D. at University of North Carolina at Chapel Hill.

Dissertation on computational stochastic predictive methodologies with spatiotemporal data.

Currently:

- Principal Scientist, Owner at SpaceTimeWorks.
- Lecturer at UCSD.

My Work ...Up Until 2 Years Ago

Analytic Tools Development for the Teradata Vantage platform

Vantage Enterprise
Open and Connected Analytics at Scale

APIs

A set of functions that enable open ecosystem access to features or data

- **Access** — REST, SQL, Python, R, Java
- **Orchestration** — QueryGrid, NOS
- **Replication** — TPT, DSA, SFTP, Native API's

In-Database Analytics

Powerful extensions to relational operations and ML extensions

- Data Transformation
- Feature Engineering, PCA
- ML modelling Functions
- Time-Series Functions
- Geospatial & Temporal Functions
- nPath Functions

CONNECTED MULTI-CLOUD DATA PLATFORM

API | SQL | PYTHON | R | JAVA

IN-DATABASE ANALYTICS | BYOA | PARTNER AI/ML

MODELOPS

ADVANCED SQL ENGINE

WORKLOAD MANAGEMENT | MACHINE LEARNING EXTENSIONS

Languages

Use preferred languages R, SQL or Python

SQL, python, R

Bring Your Own Analytics

BYOM and Script & ExecR Table Operator model training & scoring

python, pypi, R, CRAN, PMML, H2O.ai, ONNX, data iku, sas

ModelOps

Models Governance, Lifecycle Management and Monitoring extensions

Lifecycle Management

Model Deployment | Model Governance | Model Monitoring

Features Catalog

Partner Integration

Best ML/AI tools with tight integration and execution with Vantage

Spark, jupyter, Alation, R Studio, H2O.ai, Azure Machine Learning, data iku, Amazon SageMaker, DataRobot, looker, KNIME, tableau, ThoughtSpot, sas, Qlik

My Work ...6 Months Ago

Developing the Teradata Enterprise Vector Store

AI algorithms convert text into numeric representation called embeddings. They can be projected in a vector space. Words which are close in real-life are also close together in vector space.
Essentially, we are **converting unstructured data into a structured format which preserves contextual information.**

Text from internet
(unstructured)

- *People generally eat **cheese** and **butter** together in breakfast*
- ***Chocolate sauce** is used in making cake*

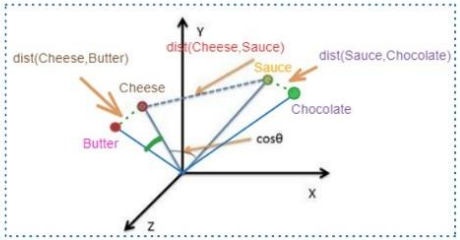
AI

Embeddings
(Numeric representation of text)

Cheese →	0.6	−0.2	0.8	0.9	−0.1	−0.9	−0.7
Butter →	0.7	0.3	0.9	−0.7	0.1	−0.5	−0.4
Sauce →	0.5	−0.4	0.7	0.8	0.9	−0.7	−0.6
Chocolate →	0.8	−0.1	0.8	−0.9	0.8	−0.5	−0.9

Dimensionality reduction of word embeddings from 7D to 2D

Vector Space
(Embeddings as vectors)



Word | Word embedding | Dimensionality reduction | Visualization of word embeddings in 2D

- A vector store **manages and organizes vector embeddings** by context
- With large datasets and multi-dimensional vectors, vector stores need to be **highly performant**
- The vector store arranges vector embeddings to be **retrievable quickly through intelligent search** (referring to standard search algorithms like vector distance, K-Means; or more efficient and advanced algorithms like Hierarchical Navigable Small World (HNSW; *keep this in mind*…)

My Work ...6 Months Ago

Developing Teradata Agentic AI

Generative AI

Transform how enterprises approach productivity, creativity and decision making.

RAG Applications

Build smart, grounded AI systems that can answer questions from documents, tickets, or proprietary databases **without hallucinating.**

Autonomous Agents

Take enterprise from reactive to proactive, as agents can plan, reason, and take multi-steps actions.

Gen AI powered Customer complaint analyzer is used to rapidly understand complaints and prepare efficient responses.

RAG application for PDF insurance contracts boost productivity and accuracy.

Augmented call-center to automate contact center human tasks.

What is this course about? Why take it?

Systems for

Scalable

Analytics

Thought Exercise!

You are the lead data engineer for a massive streaming company. You need to migrate **100 Petabytes** of raw video data from a legacy data center in New York to a new, state-of-the-art facility in San Francisco. You have two options:

a. Transfer it over a dedicated, top-of-the-line 100 Gbps (Gigabits per second) fiber-optic internet connection.
b. Load the data onto a bunch of hard drives, throw them in the back of a rented physical truck, and drive them across the country.

Which one gets the data there faster? Talk to your group, do the rough math, and cast your vote.

And the answer is……

Why?

What does 100 PB mean?

- Netflix time scale!!
- Human genome scale
- Capturing the picture of a supermassive black hole (M87*).

What's the point of this discussion?

Never underestimate the bandwidth of a station wagon full of tapes hurtling down the highway - Andrew Tanenbaum

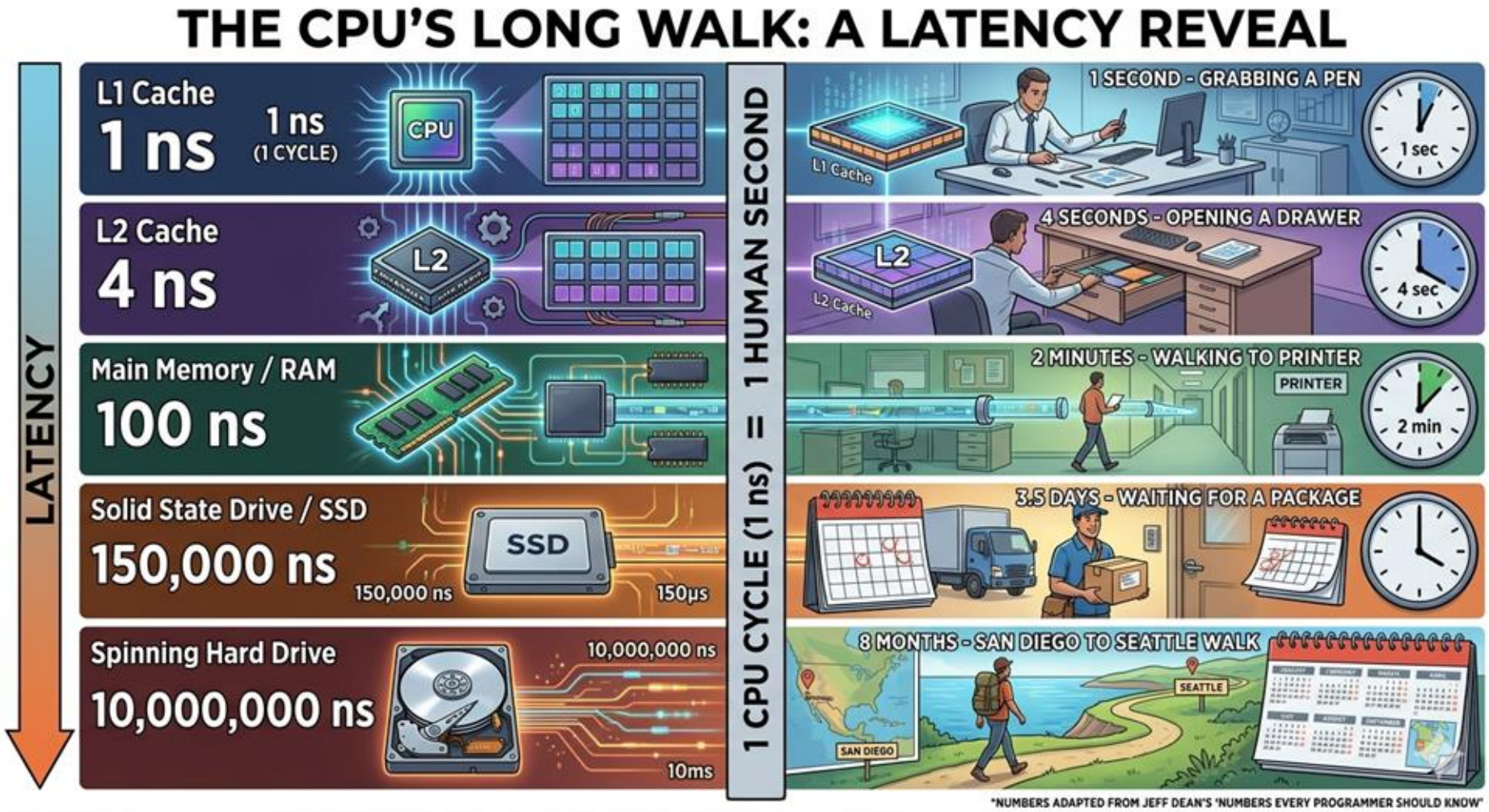
Data has mass. It is heavy, slow, and expensive to move. Therefore, you do not move the data to the compute. You move the compute to the data.

Another

*Imagine you are a modern CPU core. You are blindingly fast. To make things easy to understand, let's slow time down. Let's pretend that executing one single instruction (1 CPU cycle) takes you exactly **1 second** of human time.*

*If you need a piece of data, and it happens to be sitting right next to you in your ultra-fast L1 cache, it takes you **1 second** to grab it.*

Here is the question: If the data you need for your next calculation isn't in your cache, and you have to fetch it from the server's Main Memory (RAM), how long do you sit there waiting? What if you have to fetch it from a spinning Hard Drive? Discuss with your group.



Optimizing a machine learning algorithm's math is largely irrelevant if you don't optimize how the data gets to the processor. We are going to study In-Memory processing (Apache Spark) to keep data 'down the hall' instead of 'across the country.' We are going to study Columnar Data formats (Parquet) so that when you make that 8-month journey to the hard drive, you bring back exactly what you need, and nothing you don't.

Example: Netflix's "spot-on" suggestions

Example: Search Across Structured Data

Google

Pradeep Khosla

All News Images Videos Forums Short videos Shopping More Tools

Pradeep Khosla

Chancellor of the University of California, San Diego

YouTube • The Qualcomm Institute

Marine & Cyber-Archaeology in the Eastern Mediterranean ...

Marine & Cyber-Archaeology in the Eastern Mediterranean | **Pradeep Khosla**...

Dec 17, 2024

Place of b...

Mumbai, India

Spouse

Thespine Kavoulakis

LinkedIn

Pradeep Khosla - UC San Diego - LinkedIn

Dr. Pradeep Khosla was selected and confirmed as the eighth Chacellor of UC...

University of California San Diego
https://chancellor.ucsd.edu › about › about-the-chancellor

About Chancellor Pradeep K. Khosla

Pradeep K. **Khosla** became UC San Diego's eighth chancellor on August 1, 2012. As UC San Diego's chief executive officer, he leads a campus with more than ...

Wikipedia
https://en.wikipedia.org › wiki › Pradeep_Khosla

Pradeep Khosla

Pradeep Kumar Khosla (born March 13, 1957) is **an Indian-American computer scientist**, currently serving as the eighth chancellor of the University of ...

Overview

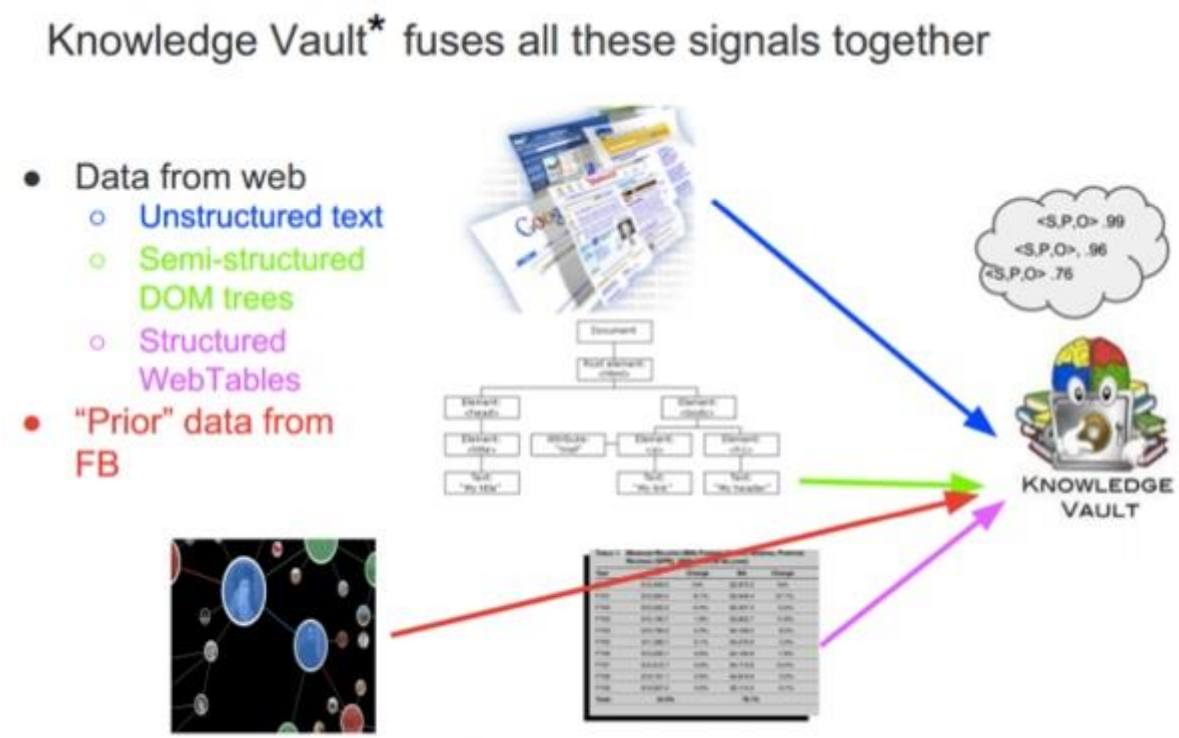
Pradeep Kumar Khosla is an Indian-American computer scientist, currently serving as the eighth chancellor of the University of California, San Diego since August 2012. He is also a former electrical engineering professor and dean at the Carnegie Mellon College of Engineering. Wikipedia

Place of birth: Mumbai, India

Spouse: Thespine Kavoulakis

Education: Carnegie Mellon University (1986), Indian Institute of Technology, Kharagpur

Large datasets + ML / DL / AI



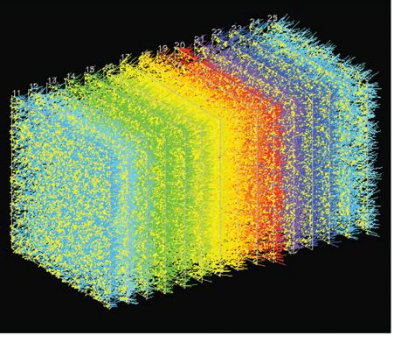
DOM:
Document Object Model

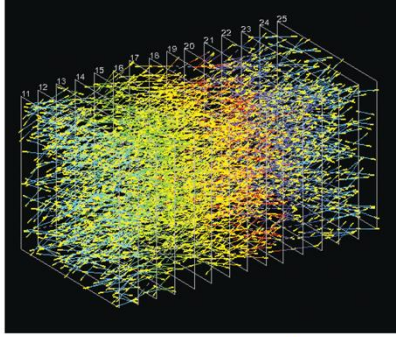
★ Details in a paper submitted to WWW'14 (Dong et al) : "Knowledge vault: A web-scale approach to probabilistic knowledge fusion". KDD '14: Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining; pages 601–610. https://doi.org/10.1145/2623330.2623623

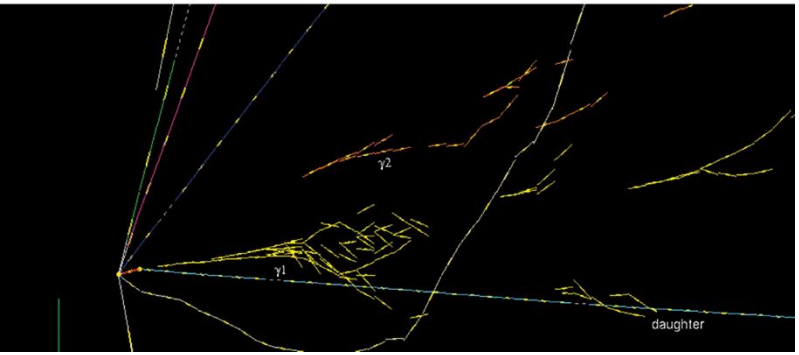
Knowledge Base Construction (KBC) process extracts tabular/relational data from large amounts of text data

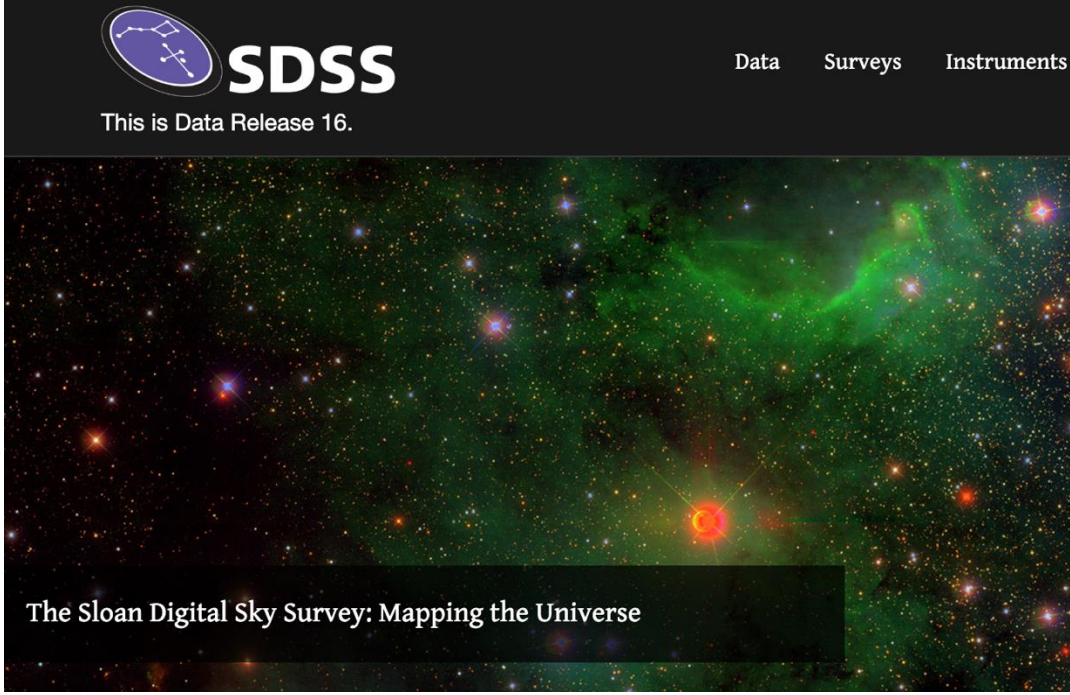
From Enterprise applications...

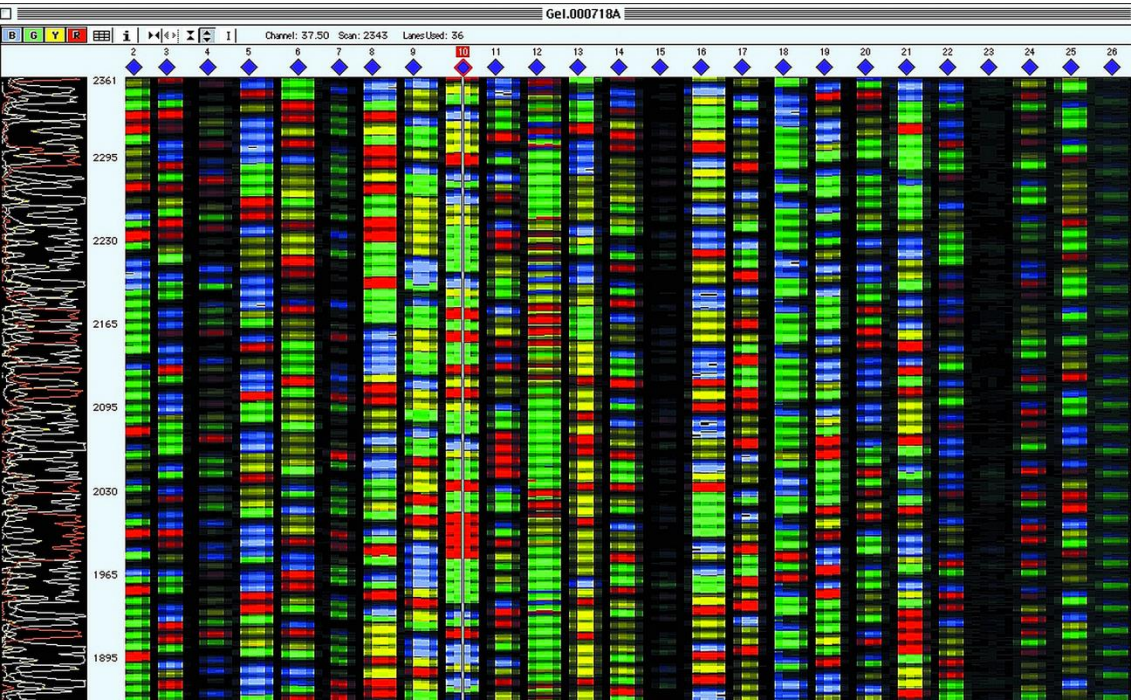
...to sciences / healthcare applications






γ2
γ1
daughter


SDSS
Data
Surveys
Instruments
This is Data Release 16.
The Sloan Digital Sky Survey: Mapping the Universe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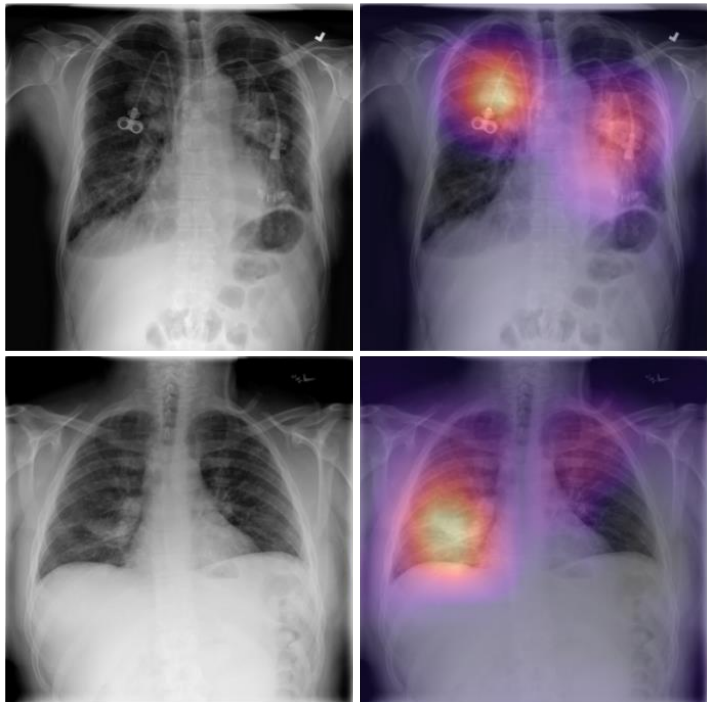
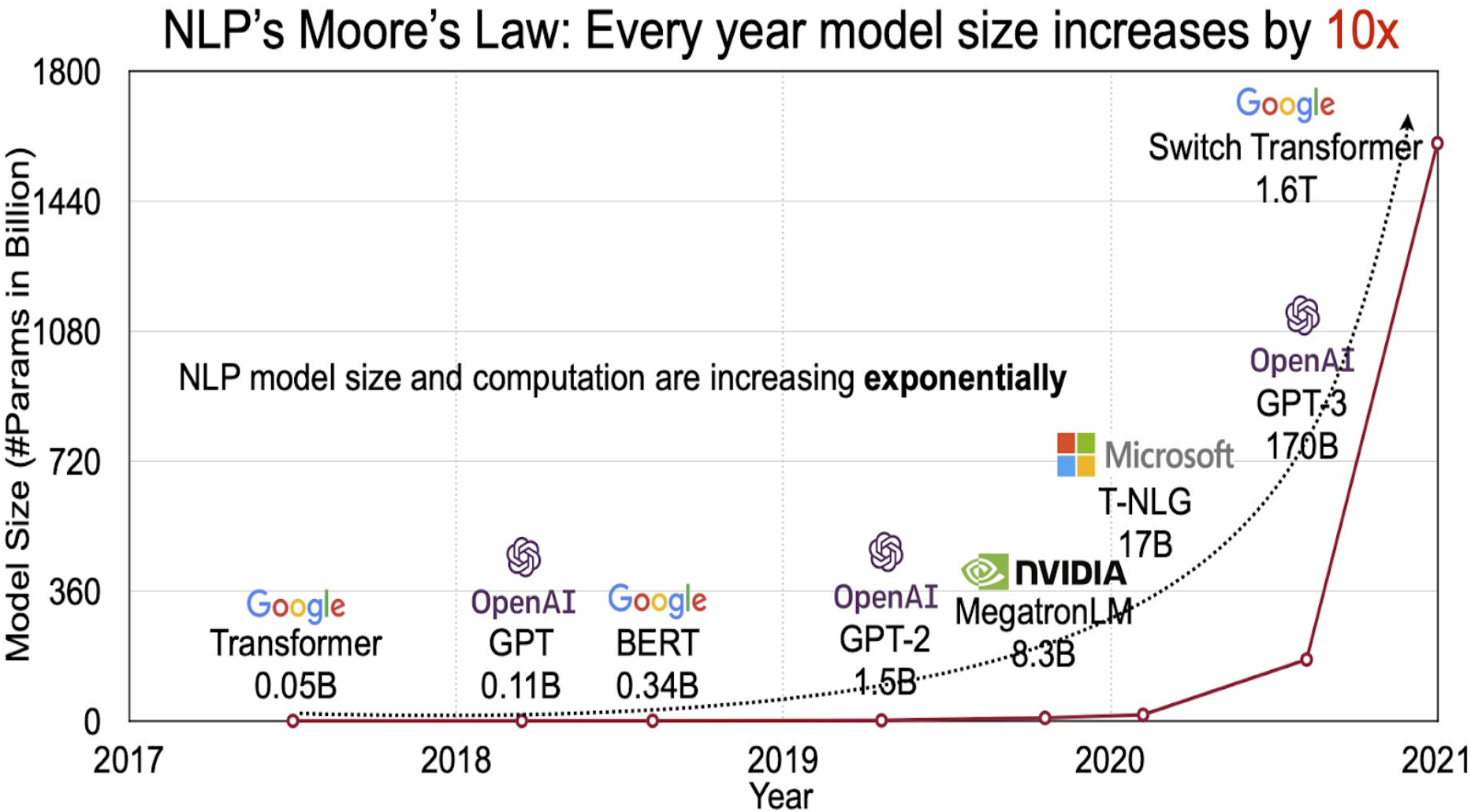


Emergence of LLMs



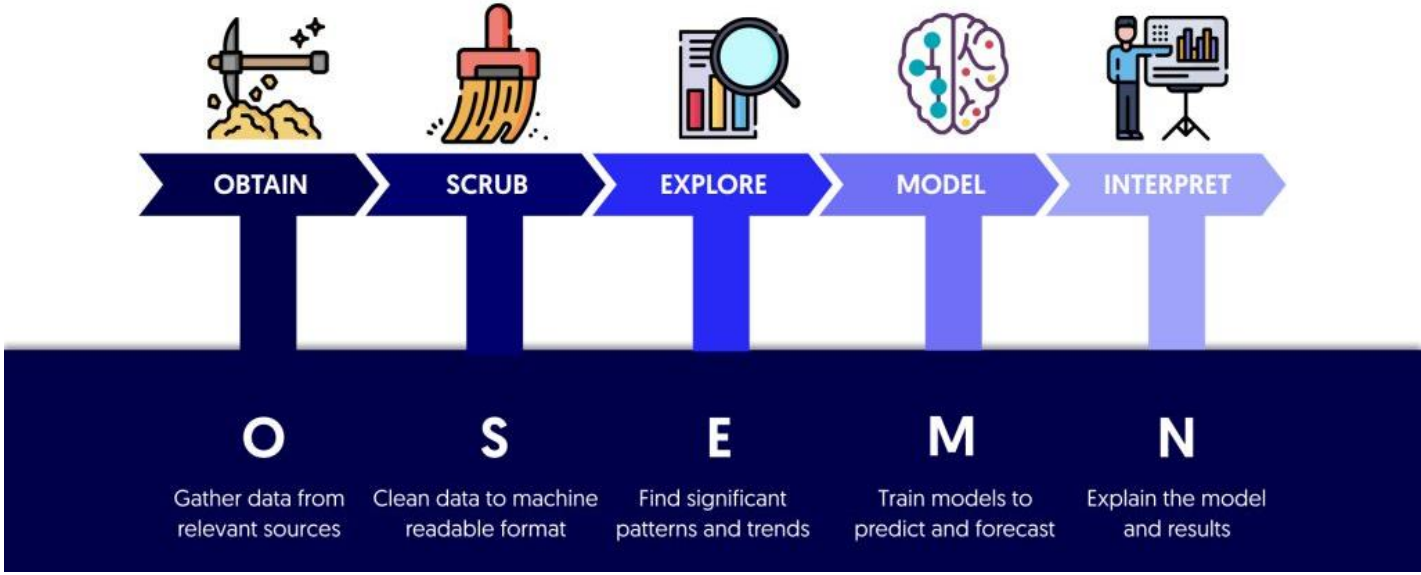
Internet Scale!

- Meta's daily log – 60 TB
- 1000 Genomes project – 200 TB
- Google's web index – 10+ PB

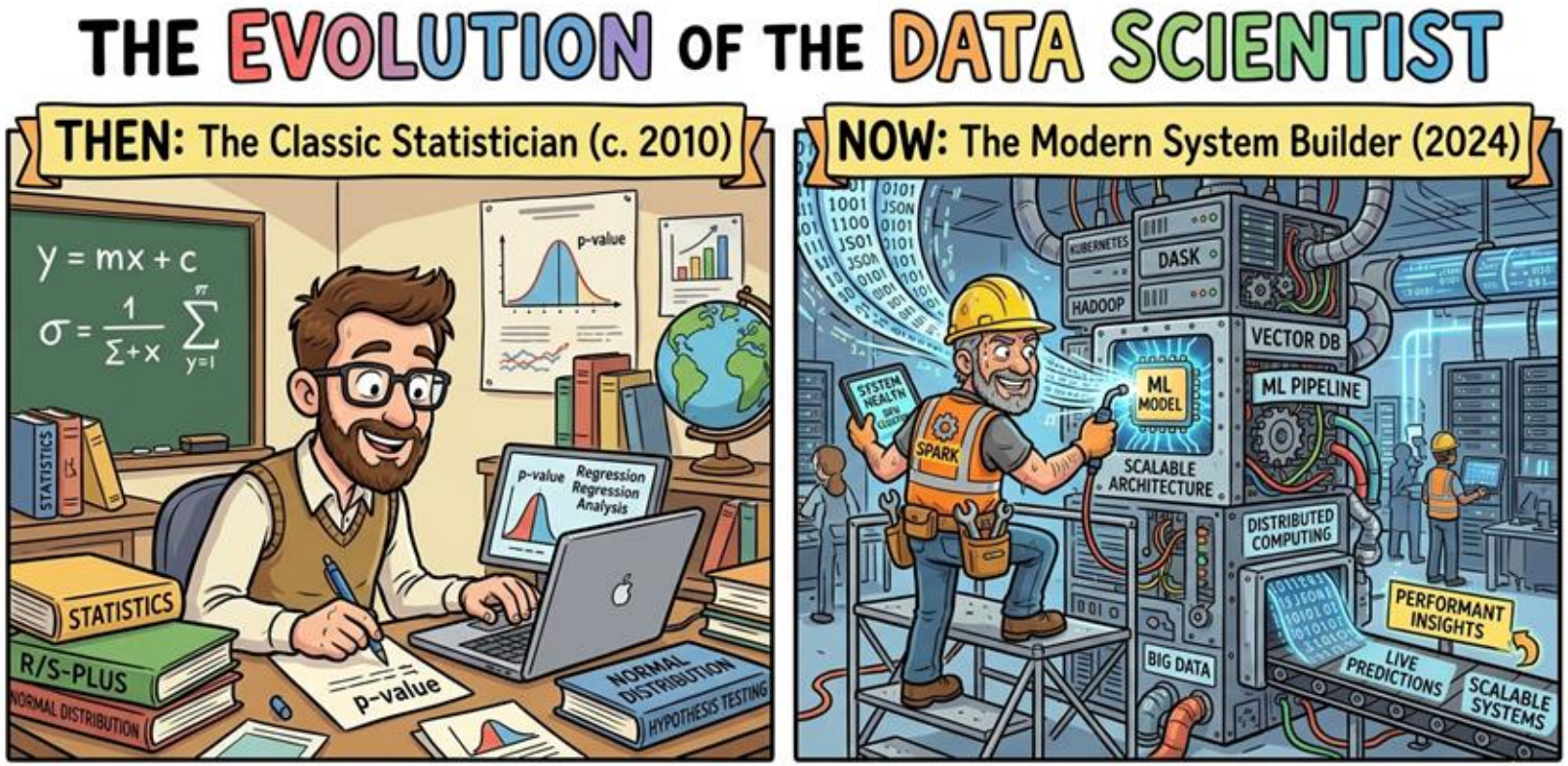
“Big Data Science”

Domain expertise:

- astronomy
- bioinformatics
- business analytics
- crime science
- digital humanities
- eco-genomics
- geography
- machine vision
- natural language processing
- seismology
- traffic engineering
- ...



Software systems for Data Analytics, ML and AI over large and complex datasets are now critical applications everywhere



Focus: p-values, sampling, bell curves, small datasets.

Focus: Big Data, Distributed Systems, Scalability, Performance, GPUs, and pipelines.

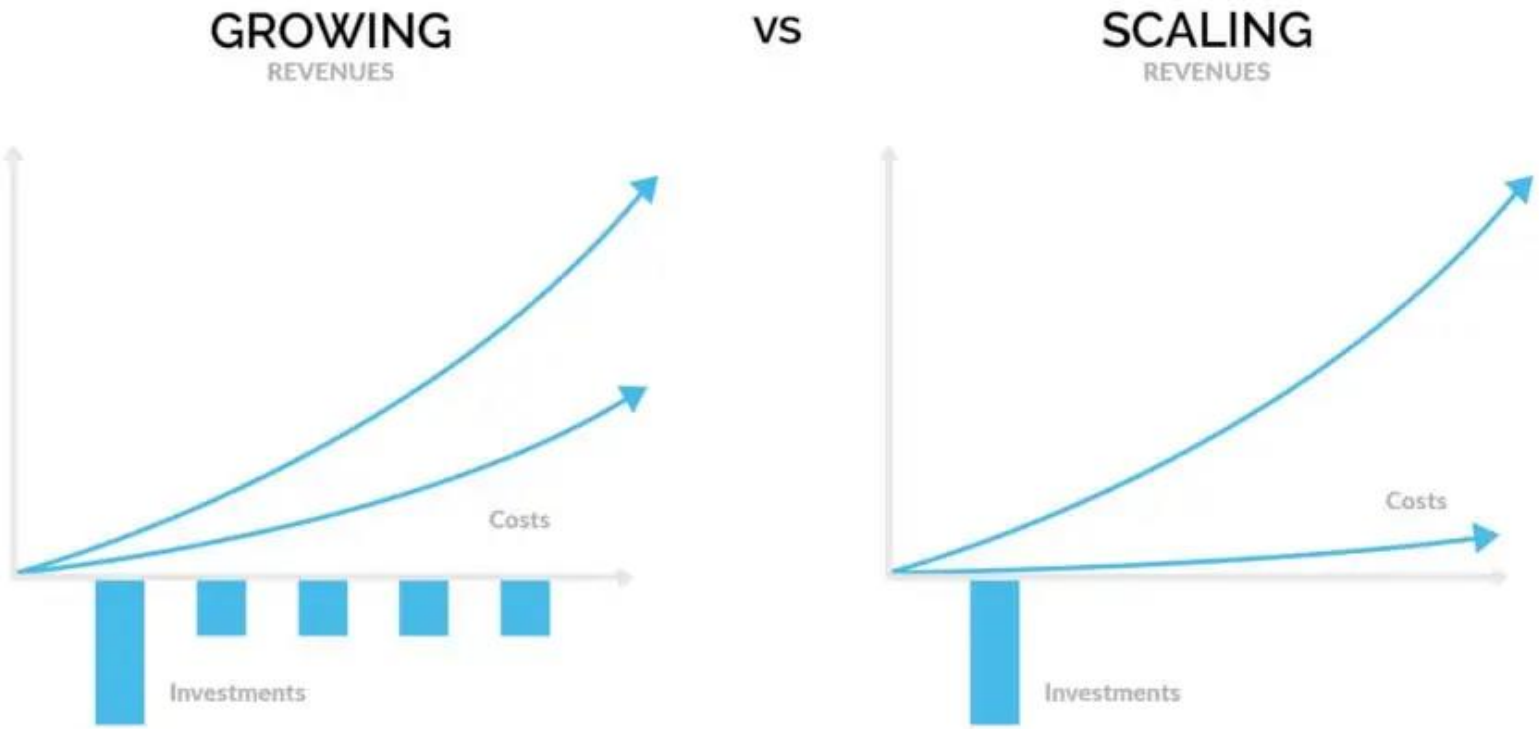
What is this course about? Why take it?

Systems for

Scalable

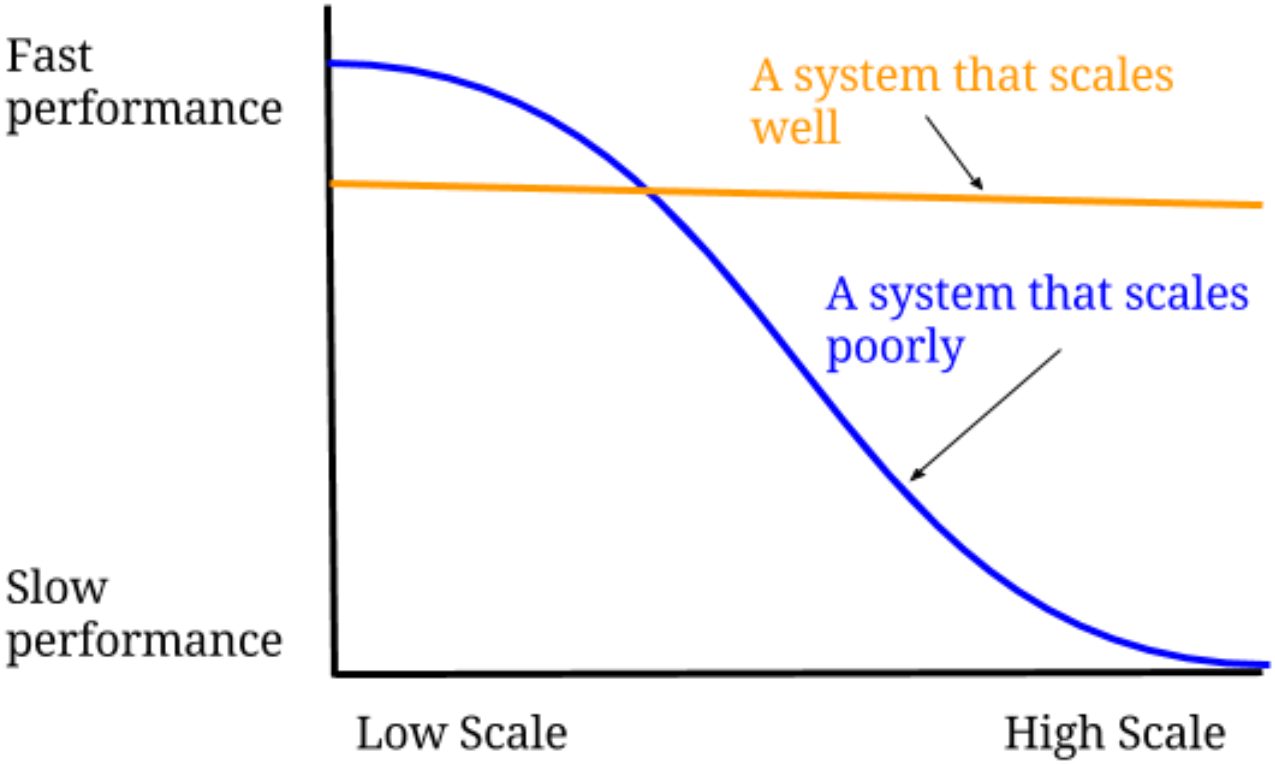
Analytics

What Is Scalability?



Business Scalability

What Is Systems Scalability?

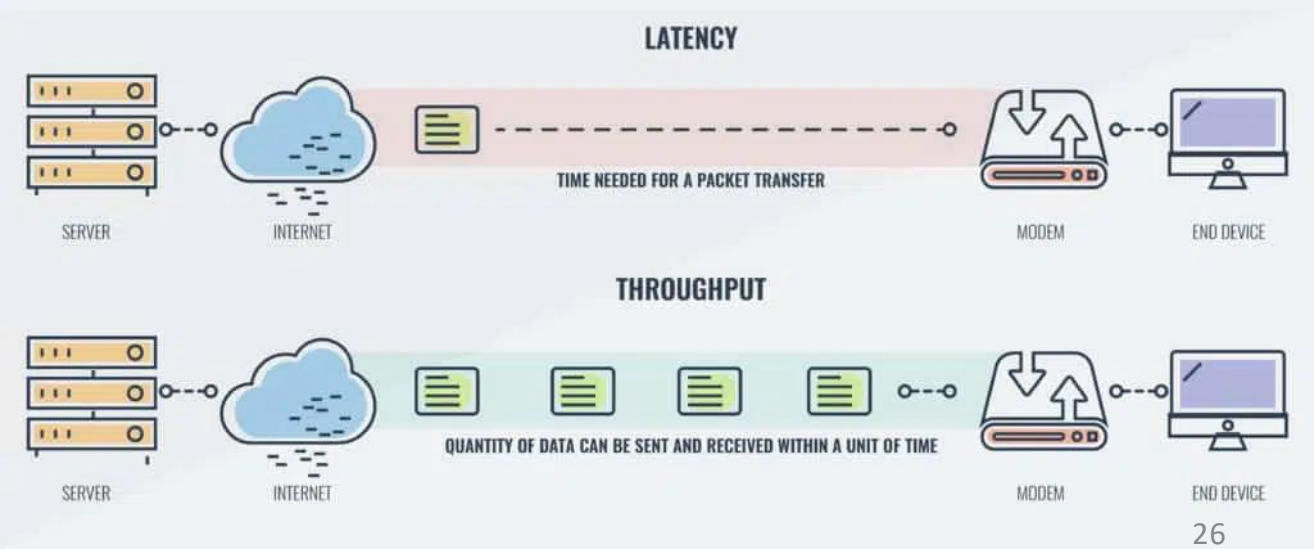


Scalability relates to performance

Measures Of System Performance

Two (primary) Measures

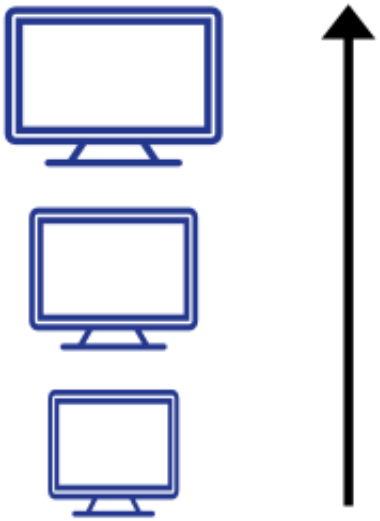
a. **Latency**
b. **Throughput**



Ways To Scale Software Systems

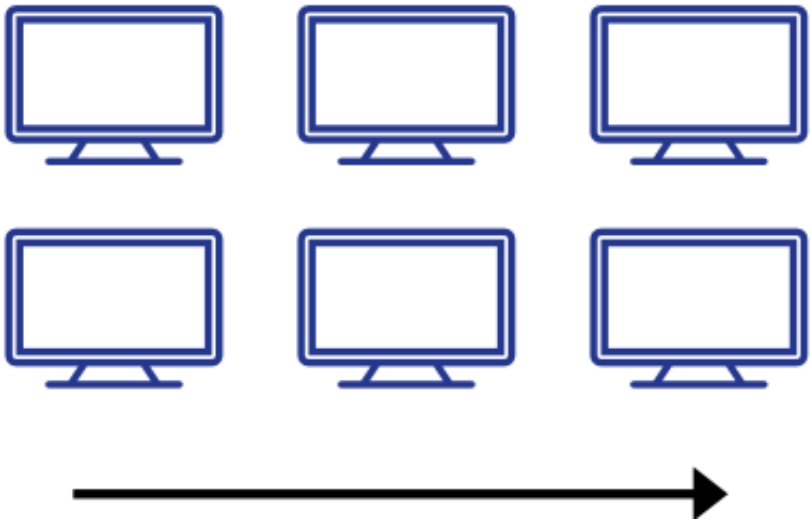
VERTICAL SCALING

Increase size of instance
(RAM, CPU etc.)



HORIZONTAL SCALING

(Add more instances)



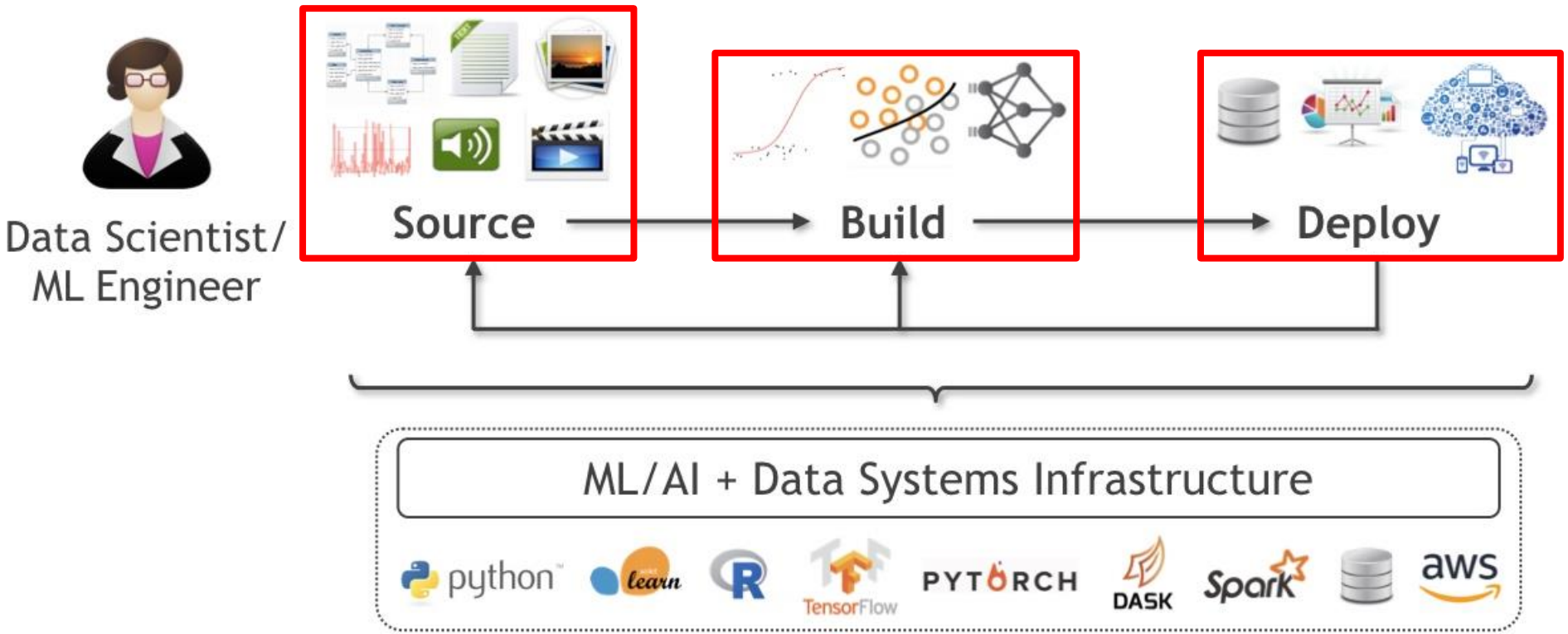
Some Questions For Scalable Design

- How many users (online and batch) will concurrently access the system ?
- How much data will the system be able to manage ?
- How many read / write operations per second does the data store need to handle ?
- What is the peak concurrency access to the system ?
- How much data can be cached to minimize the depth within the system that the requests need to travel before being responded to ?
- Can data be cached outside the system in content distribution network (CDN) to help to keep traffic away from site ?
- Is it worth caching ?
- Is data replication required for the system ? How long is it acceptable for the data synchronization to take place ?
- How much logging and events are required to the system to support the operational needs of the system, for now and future performance analysis ?
- Are there area of data contention ?
- Are the CPU intensive operations ?
- How do you plan to measure usage of the system ?
- Do you plan to meter services to throttle excessive usage ?

DSC 102 will get you thinking about the
<u>fundamentals of scalable analytics systems</u>

1. "**Systems**": What resources does a computer have? How to store and efficiently compute over large data? What is cloud?
2. "**Scalability**": How to scale and parallelize data-intensive computations?
3. **For "Analytics"**:
    1. **Source**: Data acquisition & preparation for ML
    2. **Build**: Model selection & deep learning systems
    3. **Deploying** ML models
4. Hands-on experience with scalable analytics tools

The Lifecycle of ML-based Analytics



Data acquisition
Data preparation

Feature Engineering
Training & Inference
Model Selection

Serving
Monitoring

Data Systems Concerns in ML

***Q:** How do "ML Systems" relate to ML?*

Key concerns in ML:

Accuracy

Runtime efficiency (sometimes)

Additional key *practical* concerns in ML Systems:

Scalability (and efficiency at scale)

Usability

Manageability

Developability

*Long-standing concerns in the **DB systems** world!*

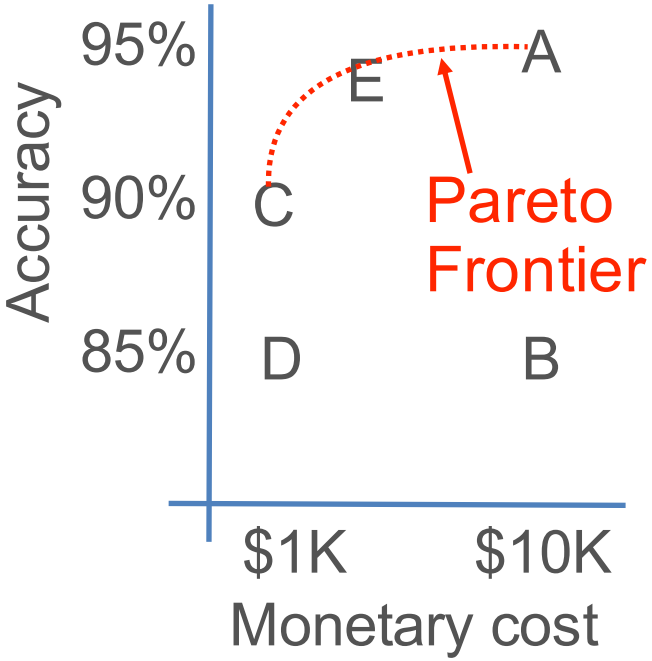
Explainability *Business requirement for DS applications*

Can often trade off <u>accuracy</u> a bit to gain on the rest!

Real-World ML: Pareto Surfaces

Q: *Suppose you are given ad click-through prediction models A, B, C, and D with accuracies of 95%, 85%, 90%, and 85%, respectively. Which one will you pick?*

Q: *What about now?*



- Real-world ML users must grapple with multi-dimensional *Pareto surfaces*: accuracy, monetary cost, training time, scalability, inference latency, tool availability, interpretability, fairness, etc.
- *Multi-objective optimization* criteria set by application needs / business policies.

Learning Outcomes of this course

- *Explain* the basic principles of the memory hierarchy, parallelism paradigms, scalable data systems, cloud computing, and containerization.
- *Identify* the abstract data access patterns of, and opportunities for parallelism and efficiency gains in, data processing and ML algorithms at scale.
- *Outline* how to use cluster and cloud services, dataflow ("Big Data") programming with MapReduce and Spark, and deep learning inference with TensorFlow and Keras.
- *Apply* the above programming skills to create end-to-end pipelines for data preparation, feature engineering, and model selection on large-scale datasets.
- *Reason* critically about practical tradeoffs between accuracy, efficiency, scalability, usability, and total cost.

What this course is NOT about

- NOT a course on databases, relational model, or SQL
  - Take DSC 100 instead (pre-requisite)
- NOT a course on internal details of RDBMSs
  - Take CSE 132C instead
- NOT a training module for how to use Spark
- NOT a course on ML or data mining *algorithmics*; instead, we focus on ML *systems*

Now for the logistics …

Prerequisites

- **DSC 100** (or equivalent) is necessary
- Transitively **DSC 80**; a mainstream ML algorithmics course is necessary
- Proficiency in Python programming
- For all other cases, email the instructor with proper justification; a waiver can be considered

Components and Grading

- **4 Programming Assignments: 40%** (8%+8%+12%+12%)
  - No late days! Plan your work well ahead.
- **Midterm Exam: 25%**
  - **Section B00: Thursday, May 7**; 6.30–7.50 pm
- **Cumulative Final Exam**: **35%**
  - **Section B00: Tuesday, June 9**; 7–10 pm
- All exams are *in-class*, *in-person* only; plan accordingly.

Grading Scheme

Hybrid of relative and absolute; grade is <u>better</u> of the two

Grade	Relative Bin (Use strictest)	Absolute Cutoff (>=)
A+	Highest 5%	95
A	Next 10% (5-15)	90
A-	Next 15% (15-30)	85
B+	Next 15% (30-45)	80
B	Next 15% (45-60)	75
B-	Next 15% (60-75)	70
C+	Next 5% (75-80)	65
C	Next 5% (80-85)	60
C-	Next 5% (85-90)	55
D	Next 5% (90-95)	50
F	Lowest 5%	< 50

Example: If score 82 but 66th percentile; Rel.: B-; Abs.: B+; so, B+

Tentative Course Schedule

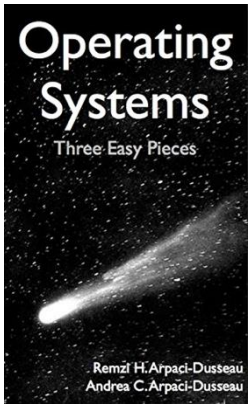
Week	Topic	
1-3	Introduction; Basics of Computer Org. and Operating Systems	Systems Principles
4	Basics of Cloud Computing	Scalability Principles
5-6	Parallel and Scalable Data Processing: Parallelism	
6	**Midterm Exam on Thursday, May 7**	
7-8	Parallel and Scalable Data Processing: Scalability	Scalable Analytics Systems
9-10	ML Data Sourcing	
	Final Exam on Tuesday, June 9	

Programming Assignments

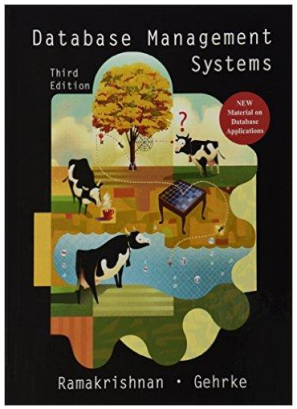
- **PA0: The SIMD & Cache Challenge**
- **PA1: Setting up AWS and Dask**
- **PA2: Data Exploration with Dask**
- **PA3: Feature Engineering and Model Selection with Spark**
- **Expectations on the PAs:**
  - Teams of 1-3; see webpage on academic integrity
  - I will cover the concepts and tools' tradeoffs in the lectures
  - TAs will explain and demo the tools; handle all Q&A
  - You are expected to put in the effort to learn the details of the tools' APIs using their documentation on your own!

Suggested Textbooks

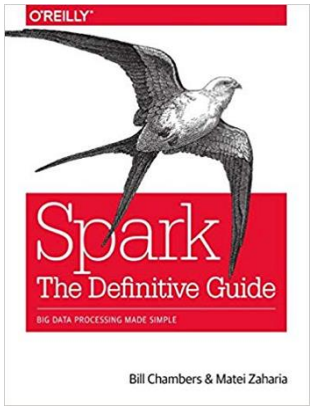
Aka “CompOrg Book”



Aka “Comet Book”



Aka “Cow Book”

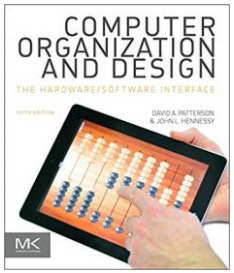


Aka “Spark Book”

(”Comet”, “Spark” books links on syllabus page; check with our library about rest.)

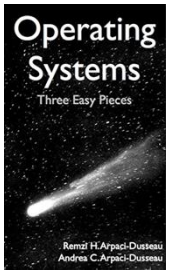
Why so many textbooks?!

Computer systems are about carefully layering *levels of abstraction*.

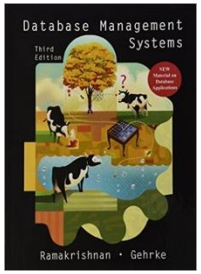


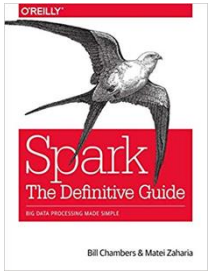
Hardware

Low-level systems software



Higher-level relational dataflows





More general scalable dataflows

Course Administrivia

- **Lectures B00**: **TTh 6.00-7.50pm PT** at **Solis 104**
  - Attendance optional but encouraged; podcast available
- **Discussions B01: M 4-4.50pm PT at Solis 104**
  - Only for talks on PAs by TAs and pre-exam prep-up.

- **Instructor: Alexander Kolovos (akolovos [at] ucsd.edu)**
  - **Office hours: Zoom meetings on Thu 4:00-5:00pm PT**
- **Canvas** for Information (Syllabus, schedule, slides, etc.)
- **Piazza** for communication, announcements, Q&A
- **Gradescope** for PA submissions

https://canvas.ucsd.edu/courses/76163

Course Staff

a. **Qiyu Li**

   OHs: Fridays 3-5pm, HDSI 336

a. **Guillame Shippee**

   OHs: Wednesdays 4-6pm, HDSI 336

Course Administrivia - PAs

- **Programming Assignments (PAs) schedule in Syllabus**
  - Strictly observed; recommended to familiarize early in each release to allow for adequate time to work on deliverables.
- **PA Groups**
  - Each group will consist of 1-3 students. Time to team up!
  - Register in a group in Canvas > People > PA_Groups by Apr 13. Groups will be locked; no switches allowed past this date. Unregistered students will be assigned into groups of 1.
  - For each PA: Single submission due per group. Each group member receives same grade earned by the joint submission.

General Dos and Do NOTs

Do:

- Follow all announcements on Piazza
- Join the lectures/discussions live AND
- Participate in discussions in class / on Piazza
- View/review videos asynchronously by yourself
- Use “DSC102 B00:” as subject prefix for emails to me/TA

Do NOT:

- Record lectures on your side without permission of instructor and other students
- Harass, intimidate, or intentionally talk over others
- Violate academic integrity on exams, or programming assignments; I am *very strict* on this matter!

Have A Fine Spring Quarter 2026!

Sailing off to new Knowledge